

MEE 201 – 2011-2012



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING

MEE 201 (MANUFACTURING TECHNOLOGY I) EXAMINATION FOR FIRST SEMESTER
2011/2012 ACADEMIC SESSION

COURSE UNIT: 2Units

INSTRUCTIONS: Answer any Four Questions

TIME ALLOWED: 2Hours

Question 1

- Explain the plastic deformation process in metal cutting operation. .
- Name the three categories of chip formed during cutting operation and write a short note on them.
- A carbide tipped cutting tool has a tool life of 20min on steel bars during a light cutting operation on a capstan lathe at 1.25m/min. Estimate the approximate tool life of this tool if the capstan operator increase the cutting speed to 3.5m/min when the cutting tool life index is one-tenth.
- Mention three ways file can be classified and three methods of filing.

Question 2

- Highlight all the tool angle functions in a single point cutting tool.
- Explain the typical properties of a cutting tool material
- Calculate the volume of metal removed when a 1inche steel bar is turned on a lathe at a feed of 0.08mm/rev with a spindle speed of 4m/min and a 3mm depth of cut.
- With the aid of an annotated diagram define riveting and state five rules governing riveting operation.

Question 3

- Explain the term machine tools and itemize 5 examples of conventional machine tools in Nigeria
- Highlight 5 safety precautions associated with both or either pneumatic or electrical machine component or tools
- With the aid of neat sketches, explain special operations that can be performed with a drilling machine.
- Mention five different types of chisel.

Question 4

- In a tabular form, list the nature of work that can be performed on the major existing machine tools.
- With the aid of diagrams, explain the sequence of operations involved with the production of bolts. State the machine tools required for its fabrication and give reasons for choosing the tool as well.
- Itemize the parts common to all machine tools in the metal workshop and explain
- Write short note on any three cutting tool material.

Question 5

- a) In a tabular form, list the mode of operation of the cutting tools and the mode of movement of workpieces in all existing machine tools.
- b) Give the reasons why automated machine tools are desirable in production line.
- c) What is the best source of power for machine tools? Explain
- d) What are the types of rake angle, explain the significance of each on cutting operation.

Question 6

- a) Mention six general properties of a carbide tool.
- b) What is the significance of coating of cutting tools
- c) Calculate the power required to drill a 10mm diameter hole in mild steel plate at a feed of 0.15mm/rev with a spindle speed of 3000rev/sec. state also the torque on the drill and the total power required from the driving motor. (Take $C = 0.65$)
- d) Mention all the cutting tool materials in manufacturing operation